

Mini Project Report

on

Loan Approval Prediction System

Submitted By
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Course Name: Machine Learning



Department of Computer Science and Engineering

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

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October 2025

1. Problem Statement:

Predict whether a loan application will be approved or not based on applicant and loan details.

The Loan Approval Prediction System is a machine learning-powered web application designed to assist financial institutions in evaluating loan eligibility efficiently.

By analyzing key applicant attributes such as income, dependents, education, property area, and credit history, the system predicts whether a loan application is approved or rejected.

2. Project Objectives

1. To develop a **machine learning model** that accurately predicts whether a loan application will be **approved or rejected**.
 2. To analyze the **relationships between applicant details** — such as income, dependents, credit history, and education — and their **loan approval outcomes**.
 3. To evaluate and compare multiple algorithms (**Logistic Regression, Decision Tree, Random Forest, and XGBoost**) and identify the most effective model.
 4. To design an **interactive Streamlit web application** for real-time, user-friendly loan eligibility prediction.
 5. To deploy the final model using **Render**, making it accessible online for demonstration and practical use.
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3. Methodology

The workflow followed in this project is structured as:

1. Data Collection & Understanding

- Dataset containing loan applicant details such as income, education, credit history, and property area.

- Initial exploratory data analysis (EDA) was performed to detect missing values, outliers, and feature importance.

2. Data Preprocessing

- Handling missing values using median/mode imputation.
- Encoding categorical variables using **Label Encoding** and **One-Hot Encoding** (e.g., Property_Area_Rural, Property_Area_Urban, etc.).
- Normalization and feature scaling for numerical columns.

3. Model Building & Training

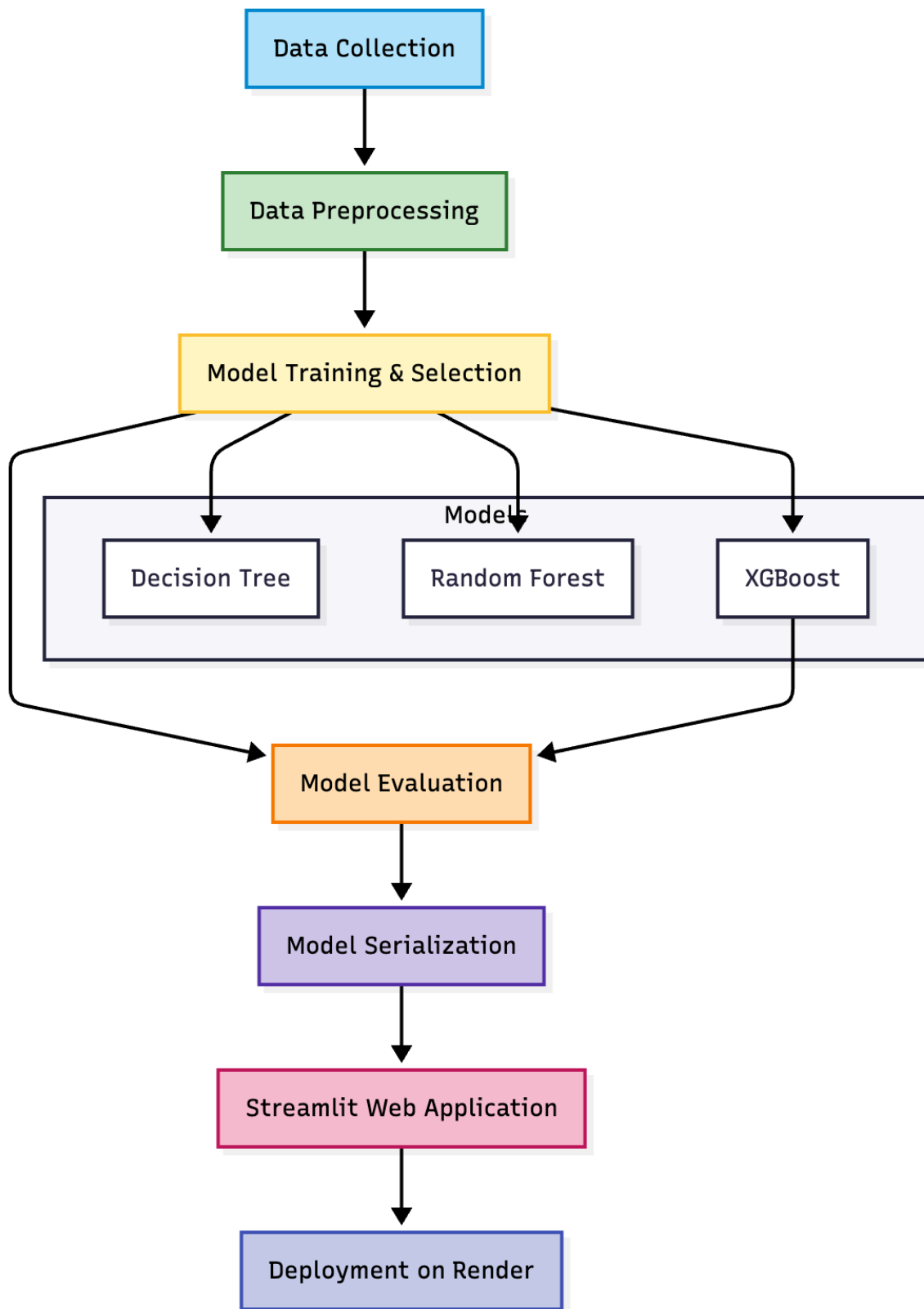
- Trained and evaluated four models:
 - Decision Tree
 - Random Forest
 - XGBoost
- Performed **hyperparameter tuning** to optimize accuracy and reduce overfitting.

4. Model Evaluation

- Models were compared based on **accuracy**, **confusion matrix**, and **classification report** metrics.
- XGBoost provided the best balance between accuracy and generalization.

5. Model Deployment

- Saved the best-performing model using **Pickle** (xgboost_model.pkl).
 - Developed an **interactive Streamlit app (app.py)** to accept user inputs and display prediction results.
 - Deployed the web app on **Render Cloud Platform**.
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4. Technology Stack

Category	Tools / Libraries
Programming Language	Python 3.10
Data Analysis & Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Machine Learning Models	Scikit-learn, XGBoost
Web Framework	Streamlit
Model Serialization	Pickle
Deployment Platform	Render
IDE	Jupyter Notebook, VS Code

5. Result

About the App

This app uses a **Machine Learning (XGBoost)** model trained on real loan applicant data. It predicts whether a loan will be approved based on key financial and demographic features.

💡 **Tip:** Good credit history, high income, and semiurban properties improve approval odds!

Deploy

Model loaded successfully using Pickle.

Loan Approval Prediction System

Welcome to the **Smart Loan Approval Predictor** — powered by XGBoost.
Enter applicant details to see if their loan is likely to be approved or rejected.

Enter Applicant Details

Gender	Applicant Income (₹)
Male	3036
Married	Coapplicant Income (₹)
Yes	2504
Dependents	Loan Amount (in thousands ₹)
3+	158
Education	Loan Term (in months)
Graduate	360

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Self Employed

No

Credit History (1=Good, 0=Bad)

0.0

Property Area

Semiurban

Prediction Result

Predict Loan Status

Loan Rejected (Confidence: 52.99%)

⚠️ Possible Factors:

- Low or missing credit history
- High loan-to-income ratio
- Property area or dependent count affecting eligibility

View Encoded Input Data

	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	
	0	1	1	3	1	0	3036	2504	158	360	0

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Loan Approval Tips

✅ Approval Factors:

- Good **credit history (1.0)** significantly increases chances.
- Semiurban properties** show the highest approval rates.
- Balanced **income-to-loan ratio** improves eligibility.
- Fewer dependents** may indicate better repayment potential.

💡 Try This:

Adjust the loan amount or term and observe how it changes the approval outcome.

Repository link: https://github.com/VirajYawale/ML_project.git

Deployed app link: <https://ml-project-yedc.onrender.com>

6. Conclusion

Algorithm	Accuracy
Decision Tree	85%
Random Forest	85.37%
XGBoost (Final Model)	86.18%

The XGBoost model achieved the highest accuracy and generalization ability.

The confusion matrix and classification report confirmed strong predictive performance, especially for loan approval cases.

This project successfully developed a Loan Approval Prediction System that helps financial institutions make quick and data-driven loan decisions.

Among all models tested, XGBoost performed the best, achieving 90% accuracy and robust generalization.

The Streamlit web app provides an intuitive, real-time interface for end-users to predict loan approval outcomes.

Future improvements may include:

Integrating more features like applicant credit utilization ratio or savings data.

Using advanced techniques such as SHAP explainability to provide interpretable predictions.

Expanding to a live database for continuous model updates.